

Final Machine Learning Project

Project Title

Using customer retail dataset implement various models and compare them.

Project Objective

The objective of this project is to build and compare multiple Machine Learning models using a customer dataset. Students will learn preprocessing, visualization, model training, prediction, evaluation, and comparison of algorithms.

Dataset Features Used

- Quantity
- UnitPrice
- Country

Machine Learning Models Used

- Logistic Regression
- Decision Tree Classifier
- K-Nearest Neighbors (KNN)

Project Workflow

1. Load customer dataset using Pandas
2. Handle missing values
3. Encode categorical columns using LabelEncoder
4. Visualize customer data using Matplotlib
5. Split data into training and testing sets
6. Train Logistic Regression model
7. Train Decision Tree model
8. Train KNN model
9. Evaluate models using Accuracy and Confusion Matrix
10. Compare model performances using graphs

Evaluation Metrics

- Accuracy Score
- Confusion Matrix

Visualization Included

- Customer Distribution Graph
- Model Accuracy Comparison Graph

Learning Outcomes

- Understand complete Machine Learning workflow
- Understand supervised learning
- Learn model evaluation techniques
- Compare different ML algorithms
- Interpret graphs and confusion matrices

Conclusion

This project demonstrates how different Machine Learning algorithms perform on the same dataset. Students learn the importance of preprocessing, feature selection, evaluation, and model comparison in real-world Machine Learning applications.